

ARRANGE SPATIAL DATA, NETWORKS AND TREES

Bùi Tiến Lên

01/01/2020



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

Contents



1. Arrange Spatial Data

2. Arrange Networks and Trees



Arrange Spatial Data

- Geometry
- Scalar Fields: One Value
- Vector Fields: Multiple Values
- Tensor Fields: Many Values

The Big Picture



Arrange Spatial Data

→ Use Given

→ Geometry

→ *Geographic*

→ *Other Derived*



→ Spatial Fields

→ *Scalar Fields (one value per cell)*

→ *Isocontours*

→ *Direct Volume Rendering*



→ *Vector and Tensor Fields (many values per cell)*

→ *Flow Glyphs (local)*

→ *Geometric (sparse seeds)*

→ *Textures (dense seeds)*

→ *Features (globally derived)*





Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- Geometric data does not necessarily have attributes associated with it: it conveys shape information directly through the spatial position of its elements.
- A common source of derived geometry data is **geographic** information about the Earth.
- Geometry is also frequently derived from computations on spatial fields.



Idiom: choropleth map

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- A **choropleth** map shows a quantitative attribute encoded as color over regions delimited as area marks, where the shape of each region is determined by using given geometry.

Idiom	Choropleth Map
What: Data	Geographic geometry data. Table with one quantitative attribute per region.
How: Encode	Space: use given geometry for area mark boundaries. Color: sequential segmented colormap

- small regions are less visually salient

Idiom: choropleth map (cont.)



Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

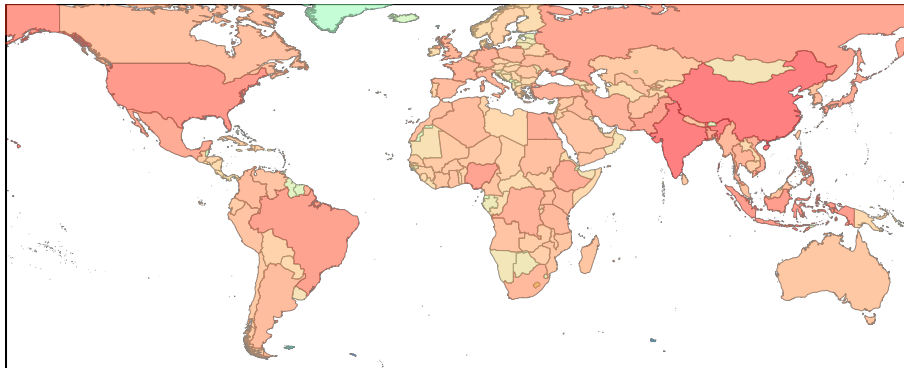
Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks





Idiom: choropleth map (cont.)

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange

Networks and Trees

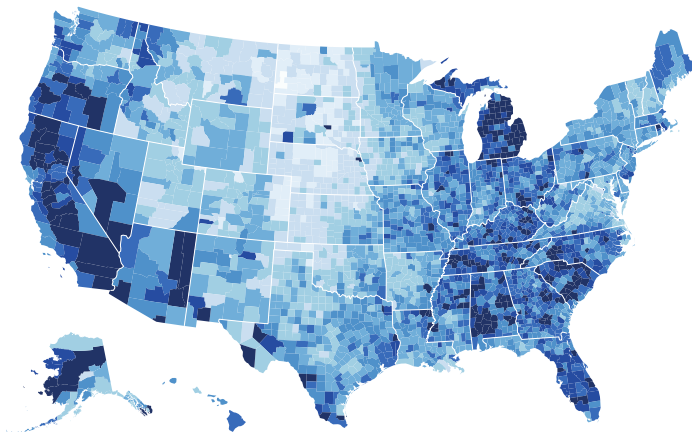
Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- Choropleth map showing regions as area marks using given geometry, where a quantitative attribute is encoded with color.





Scalar Fields: One Value

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- A scalar spatial field has a single value associated with each spatially defined cell.
- Scalar fields are often collected through medical imaging, where the measured value is radio-opacity in the case of computed tomography (CT) scans and proton density in the case of magnetic resonance imaging (MRI) scans.
- There are three major families of idioms for visually encoding scalar fields: **slicing**, **isocontours** and **direct volume rendering**



Scalar Fields: One Value (cont.)

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

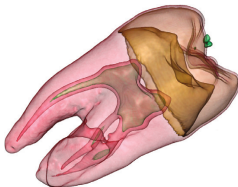
Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- Spatial scalar fields shown with three different idioms. (a) A single 2D slice of a turbine blade dataset. (b) Multiple semitransparent isosurfaces of a 3D tooth dataset. (c) Direct volume rendering of the entire 3D turbine dataset.



(a)



(b)



(c)



Isocontours

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- A set of **isolines**, namely, lines that represent the contours of a particular level of the scalar value, can be derived from a scalar spatial field
- Color coding the regions between the contours with a sequential colormap yields a **contour plot**.



Idiom: Topographic Terrain Map

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume

Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange

Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

Idiom	Topographic Terrain Map
What: Data What: Derived How: Encode Why: Tasks Scale	2D spatial field; geographic data. Geometry: set of isolines computed from field. Use given geographic data geometry of points, lines and region marks. Use derived geometry as line marks (blue). Query shape. Dozens of contour levels.

Idiom: Topographic Terrain Map (cont.)



Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

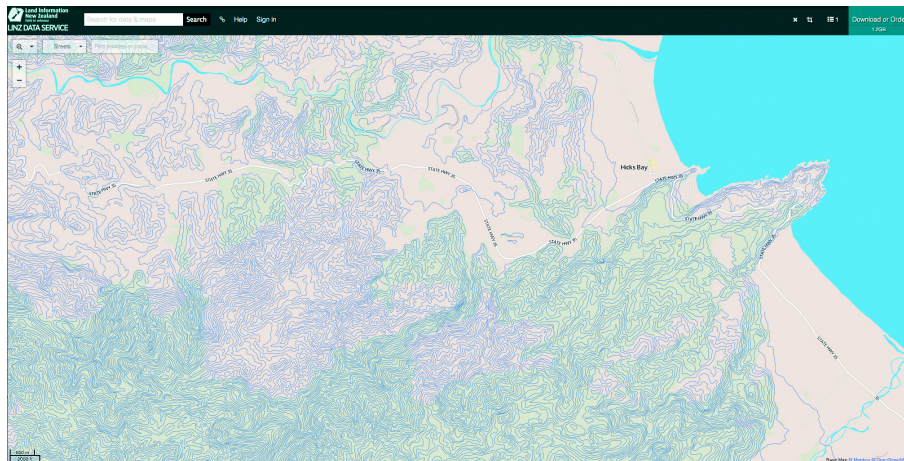
Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- Topographic terrain map, with isolines in blue.





Idiom: isosurfaces

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange

Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- The idiom of **isosurfaces** transforms a 3D scalar spatial field into one or more derived 2D surfaces that represent the contours of a particular level of the scalar value

Idiom	Flexible Isosurfaces
What: Data What: Derived How: Encode Why: Tasks Scale	Spatial field. Geometry: surfaces. Tree: simplified contour tree. Surfaces: use given. Tree: line marks, vertical spatial position encodes isovalue. Query shape. One dozen contour levels.



Idiom: isosurfaces (cont.)

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

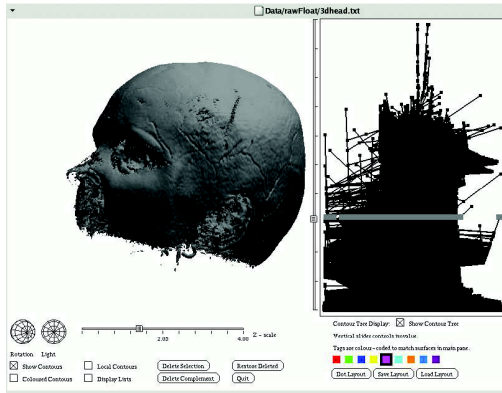
Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- A full contour tree with over 1.5 million edges does not help the user explore isosurfaces.



Idiom: isosurfaces (cont.)



Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

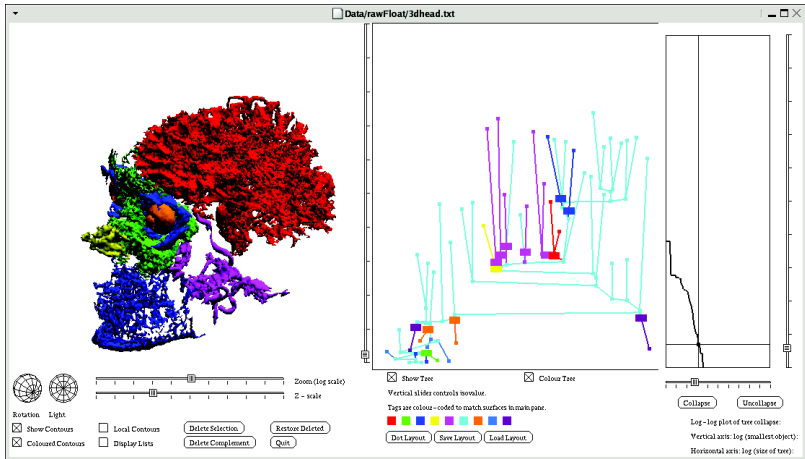
Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- The flexible isosurfaces idiom uses the simplified contour tree of under 100 edges to help users identify meaningful structure.





Direct Volume Rendering

Geometry

Scalar Fields: One
Value

Isocontours

**Direct Volume
Rendering**

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- The **direct volume rendering** idiom creates an image directly from the information contained within the scalar spatial field, without deriving an intermediate geometric representation of a surface.
- A crucial visual encoding design choice with direct volume rendering is picking the **transfer function** that maps changes in the scalar value to opacity and color.

Arrange Spatial Data

Geometry

Scalar Fields: One Value

Isocontours

Direct Volume Rendering

Vector Fields: Multiple Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many Values

Arrange Networks and Trees

Connection: Link Marks

Matrix Views

Costs and Benefits: Connection versus Matrix

Containment: Hierarchy Marks

Idiom: Multidimensional Transfer Functions

Idiom	Multidimensional Transfer Functions
What: Data	3D spatial field.
What: Derived	3D spatial field: gradient of original field.
What: Derived	Table: two key attributes, values binned from min to max for both data and derived data. One derived quantitative value attribute (item count per bin).
How: Encode	3D view: use given spatial field data, color and opacity from multidimensional transfer function. Joint histogram view: area marks in 2D matrix alignment, grayscale sequential colormap.

18



Example

Geometry

Scalar Fields: One
Value

Isocontours

**Direct Volume
Rendering**

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

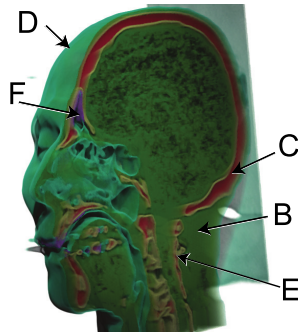
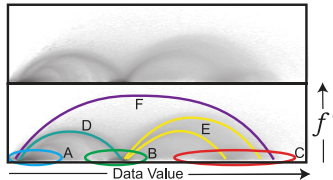
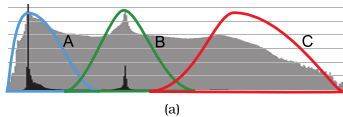
Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

Simian allows users to construct multidimensional transfer functions for direct volume rendering using a derived space. (a) The standard 1D histogram can show the three basic materials: (A) air, (B) soft tissue, and (C) bone. (b) The full 2D derived space allows material boundaries to be distinguished as well. (c) Volume rendering of head dataset using the resulting 2D transfer function, showing material boundaries of (D) air–tissue, (E) tissue–bone, and (F) air–bone.





Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

**Vector Fields: Multiple
Values**

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

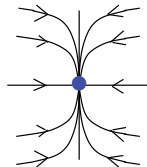
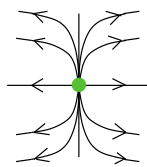
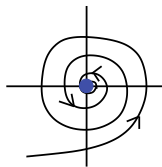
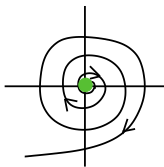
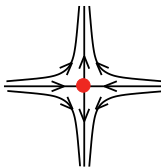
Containment:
Hierarchy Marks

- Vector field datasets are often associated with the application domain of computational fluid dynamics (CFD), as the outcome of flow simulations or measurements.
- Flow vis in particular deals with a specific kind of vector field, a velocity field, that contains information about both direction and magnitude at each cell.
- Time-varying flow datasets are called **unsteady**, as opposed to **steady** flows where the behavior does not change over time.
- There are four major families of vector field spatial visual encoding idioms: *flow glyph*, *geometric flow*, *texture flow* and *feature flow*



Vector Fields (cont.)

- One of the features of interest in flows are the **critical points**, the points in a flow field where the velocity vanishes (**sources**, **sinks** and **saddle points**)





Flow Glyphs

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

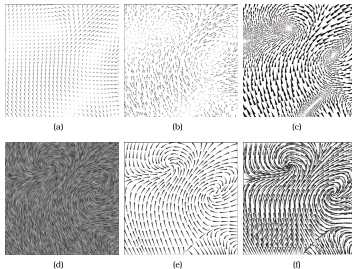
Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- The **flow glyph** idioms show local information about a cell in the field using an object with internal substructure; one of the most basic choices is an arrow
- An empirical study compared human response to six different 2D flow vis idioms. (a) arrow glyphs on a regular grid. (b) arrow glyphs on a jittered grid. (c) triangular wedge glyphs inspired by oil painting strokes. (d) dense texturebased Line Integral Convolution (LIC). (e) curved arrow glyphs with image-guided streamline seeding. (f) curved arrow glyphs with regular grid streamline seeding.





Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- The **geometric flow** idioms compute derived geometric data from the original field using trajectories computed from a sparse set of seed points and then directly show the derived geometry.
- The trajectory that a specific particle will follow is called a **streamline** for a steady field and a **pathline** for an unsteady (time-varying) field.
- A **timeline** is formed by connecting a front of pathlines over time.



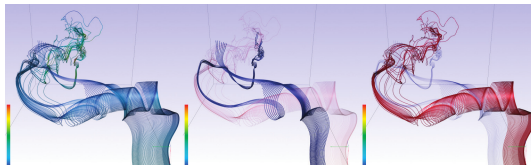
Idiom: Similarity-Clustered Streamlines

Idiom	Similarity-Clustered Streamlines
What: Data	Spatial field: 3D vector field.
What: Derived	Geometry: streamlines or pathlines.
What: Derived	One attribute per streamline/pathline (signature).
What: Derived	Cluster hierarchy of streamlines/pathlines.
How: Encode	Use derived geometry of lines, color, and opacity according to cluster.
Why: Tasks	Find features, query shape.
Scale	Field: millions of samples. Geometry: hundreds of streamlines.

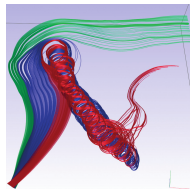
Example



- Geometric flow vis idioms showing a sparse set of particle trajectories, with seeding and coloring according to similarity. (a) Streamlines: all clusters equally opaque; purple cluster emphasized; red cluster emphasized. (b) Pathlines, colored by three clusters.



(a)



(b)



Texture Flow

- The **texture flow** idioms also rely on particle tracing, but with dense coverage across the entire field rather than from a carefully selected set of seed points.
- They are most commonly used for 2D fields or fields on 2D surfaces.



Feature Flow

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- The **feature flow** vis idioms rely on global computations across the entire vector field to explicitly locate all instances of specific structures of interest, such as critical points, vortices, and shock waves.



Tensor Fields

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

**Tensor Fields: Many
Values**

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- Tensor fields typically contain a matrix at each cell in the field, capturing more complex structure than what can be expressed in a vector field.
- Tensor fields can measure properties such as stress, conductivity, curvature, and diffusivity.
- All of the idiom families used for vector fields are also used for tensor fields: local *glyphs*, sparse *geometry*, dense *textures*, and explicitly derived *features*.



Idiom: Ellipsoid Tensor Glyphs

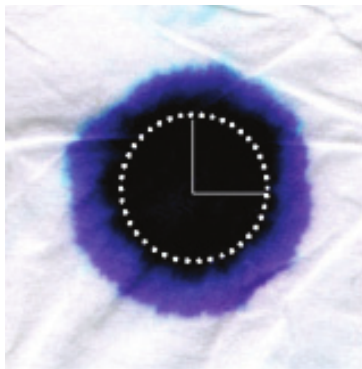
- Tensor quantities can be naturally decomposed into orientation and shape information; these quantities can be visually encoded with a 3D glyph.
- A shape may be **isotropic**, where each direction is the same, or **anisotropic**, where there is a directional asymmetry

Idiom	Ellipsoid Tensor Glyphs
What: Data What: Derived How: Encode	Spatial field: 3D tensor field. Three quantitative attributes: tensor shape. Three vectors: tensor orientation. Glyph showing six derived attributes, color and opacity according to cluster.

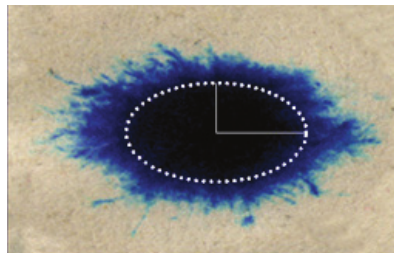
Example



- 2D diffusion illustrated with ink and paper. (a) Isotropic Kleenex. (b) Anisotropic newspaper



(a)



(b)

Example



Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

**Tensor Fields: Many
Values**

Arrange
Networks and
Trees

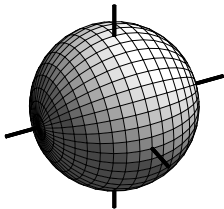
Connection: Link
Marks

Matrix Views

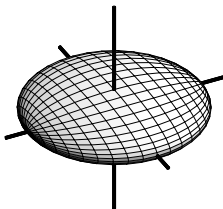
Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

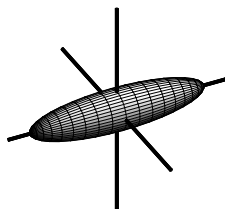
- Ellipsoid glyphs can show three basic shapes. (a) Isotropic: sphere. (b) Partially anisotropic: planar.



(a)



(b)



(c)



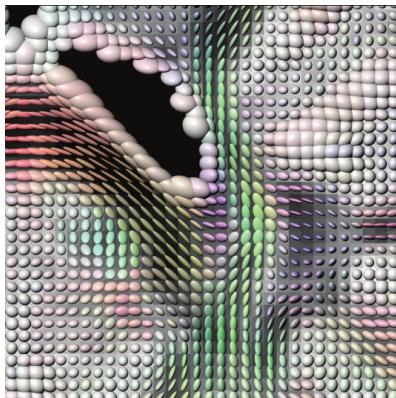
Example

Geometry
Scalar Fields: One
Value
Isocontours
Direct Volume
Rendering
Vector Fields: Multiple
Values
Flow Glyphs
Geometric Flow
Texture Flow
Feature Flow
**Tensor Fields: Many
Values**

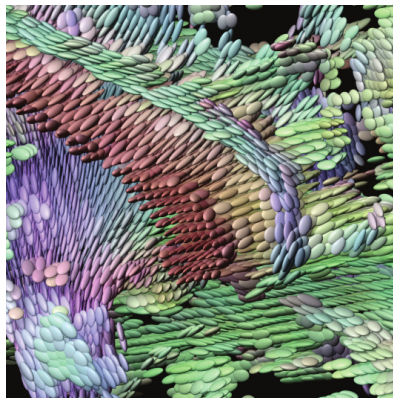
Arrange
Networks and
Trees

Connection: Link
Marks
Matrix Views
Costs and Benefits:
Connection versus
Matrix
Containment:
Hierarchy Marks

- Ellipsoid glyphs show shape and orientation of tensors at each cell in a field.
(a) 2D slice. (b) 3D field, with isotropic glyphs filtered out.



(a)



(b)



Arrange Networks and Trees

- Connection: Link Marks
- Matrix Views
- Costs and Benefits: Connection versus Matrix
- Containment: Hierarchy Marks



The Big Picture

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume

Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

Arrange Networks and Trees



Node-Link Diagrams Connection Marks

✓ NETWORKS

✓ TREES



Adjacency Matrix Derived Table

✓ NETWORKS

✓ TREES



Enclosure Containment Marks

✗ NETWORKS

✓ TREES





Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- The most common visual encoding idiom for tree and network data is with **node–link diagrams**, where nodes are drawn as point marks and the links connecting them are drawn as line marks
- This idiom uses connection marks to indicate the relationships between items.



Layout

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

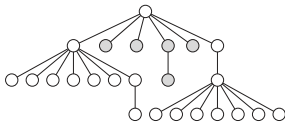
Connection: Link
Marks

Matrix Views

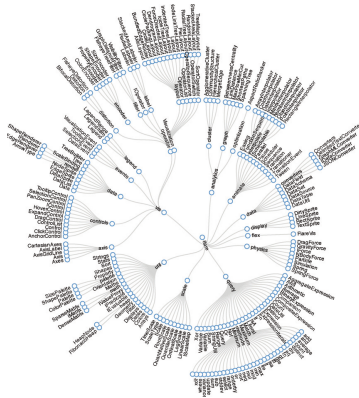
Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- Node-link layouts of small trees. (a) Triangular vertical for tiny tree. (b) Spline radial layout for small tree



(a)



(b)



Layout (cont.)

Geometry
Scalar Fields: One
Value
Isocontours
Direct Volume
Rendering
Vector Fields: Multiple
Values
Flow Glyphs
Geometric Flow
Texture Flow
Feature Flow
Tensor Fields: Many
Values

Arrange Networks and Trees

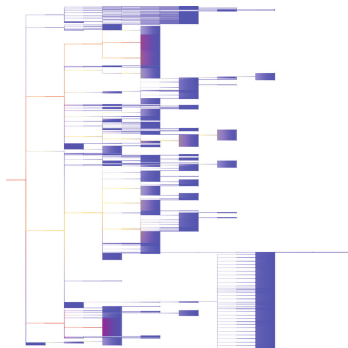
Connection: Link
Marks

Matrix Views

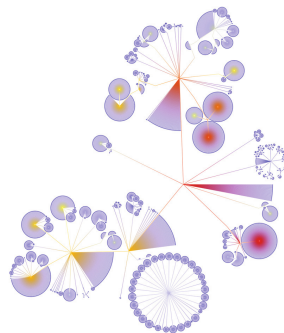
Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- Two layouts of a 5161-node tree. (a) Rectangular horizontal node-link layout. (b) BubbleTree node-link layout.



(a)



(b)



Example: Force-Directed Placement

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange

Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- One of the most widely used idioms for node–link network layout using connection marks is **force-directed placement**.
- The algorithms are designed to minimize the number of distracting artifacts such as edge crossings and node overlaps, so the spatial location of the elements is a side effect of the computation rather than directly encoding attributes.
- One weakness of force-directed placement is that the layouts are often **nondeterministic**



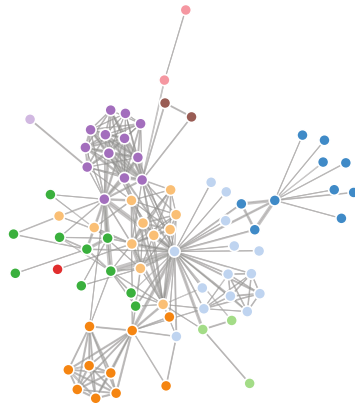
Idiom: Force-Directed Placement

Idiom	Force-Directed Placement
What: Data How: Encode Why: Tasks Scale	Network. Point marks for nodes, connection marks for links. Explore topology, locate paths. Nodes: dozens/hundreds. Links: hundreds. Node/link density: $L < 4N$.

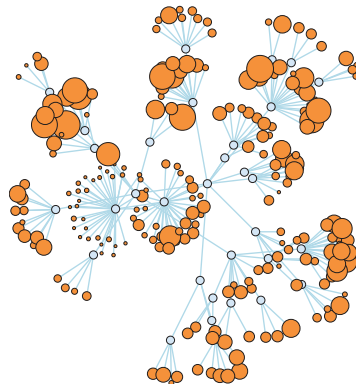
Idiom: Force-Directed Placement (cont.)



- Node-link layouts of small networks. (a) Force-directed placement of small network of 75 nodes, with size coding for link attributes. (b) Larger network, with size coding for node attributes.



(a)



(b)



Multilevel Network

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- Many recent approaches to scalable network drawing are **multilevel network idioms**, where the original network is augmented with a derived cluster hierarchy to form a compound network.
- The cluster hierarchy is computed by **coarsening** the original network into successively simpler networks that nevertheless attempt to capture the most essential aspects of the original's structure.
- By laying out the simplest version of the networks first, and then improving the layout with the more and more complex versions, both the speed and quality of the layout can be improved.



Idiom: sfdp

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume

Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:

Connection versus
Matrix

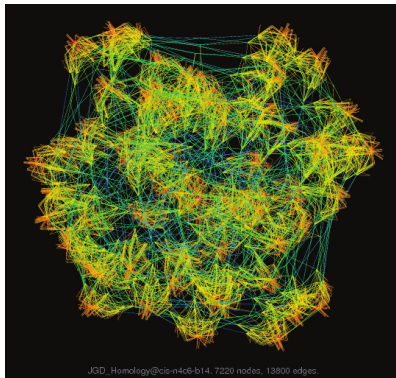
Containment:
Hierarchy Marks

Idiom	Multilevel Force-Directed Placement (sfdp)
What: Data What: Derived What: Encode Why: Tasks Scale	Network. Cluster hierarchy atop original network. Point marks for nodes, connection marks for links. Explore topology, locate paths and clusters. Nodes: 1000–10,000. Links: 1000–10,000. Node/link density: $L < 4N$.

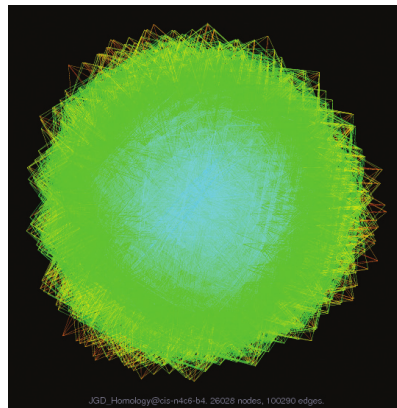


Idiom: sdfp (cont.)

- Multilevel graph drawing with sdfp. (a) Cluster structure is visible for a large network of 7220 nodes and 13,800 edges. (b) A huge graph of 26,028 nodes and 100,290 edges is a “hairball” without much visible structure.



(a)



(b)



Example: Adjacency Matrix View

Geometry
Scalar Fields: One
Value
Isocontours
Direct Volume
Rendering
Vector Fields: Multiple
Values
Flow Glyphs
Geometric Flow
Texture Flow
Feature Flow
Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks
Matrix Views
Costs and Benefits:
Connection versus
Matrix
Containment:
Hierarchy Marks

- Network data can also be encoded with a matrix view by deriving a table from the original network data.
- A network can be visually encoded as an **adjacency matrix** view, where all of the nodes in the network are laid out along the vertical and horizontal edges of a square region and links between two nodes are indicated by coloring an area mark in the cell in the matrix that is the intersection between their row and column.



Idiom

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume

Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:

Connection versus
Matrix

Containment:
Hierarchy Marks

Idiom	Adjacency Matrix View
What: Data What: Derived How: Encode Scale	Network. Table: network nodes as keys, link status between two nodes as values. Area marks in 2D matrix alignment. Nodes: 1000. Links: one million.



Idiom (cont.)

Geometry
Scalar Fields: One
Value
Isocontours
Direct Volume
Rendering
Vector Fields: Multiple
Values
Flow Glyphs
Geometric Flow
Texture Flow
Feature Flow
Tensor Fields: Many
Values

Arrange Networks and Trees

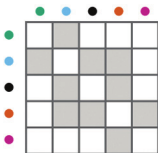
Connection: Link
Marks

Matrix Views

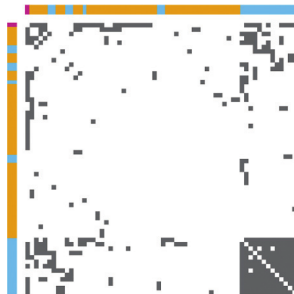
Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

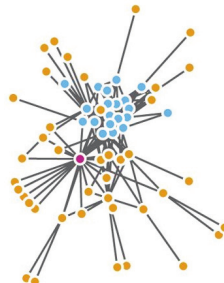
- Comparing node-link matrix and matrix views of a network. (a) Node-link and matrix views of small network. (b) Matrix view of larger network. (c) Node-link view of larger network.



(a)



(b)



(c)



Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange
Networks and
Trees

Connection: Link
Marks

Matrix Views

**Costs and Benefits:
Connection versus
Matrix**

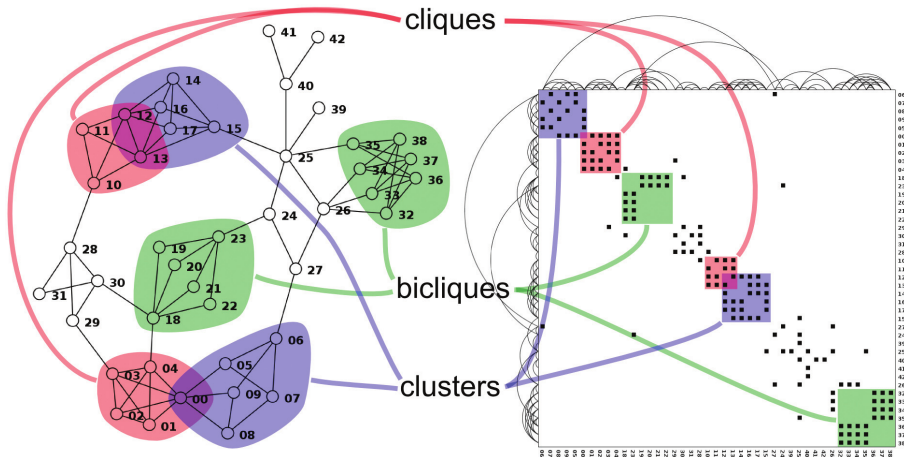
Containment:
Hierarchy Marks

- The great strength of node–link layouts is that for sufficiently small networks they are extremely intuitive for supporting many of the abstract tasks that pertain to network data.
- Their weakness is that past a certain limit of network size and link density, they become impossible to read because of occlusion from edges crossing each other and crossing underneath nodes.
- A major strength of matrix views is perceptual scalability for both large and dense networks.
- Another strength of matrix views is their predictability, stability, and support for reordering.
- One major weakness of matrix views is unfamiliarity



Costs and Benefits (cont.)

- Characteristic patterns in matrix views and node-link views: both can show cliques and clusters clearly





Example: Treemaps

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

- The idiom of **treemaps** is an alternative to node–link tree drawings, where the hierarchical relationships are shown with containment rather than connection.
- All of the children of a tree node are enclosed within the area allocated that node, creating a nested layout. The size of the nodes is mapped to some attribute of the node.



Idiom: Treemaps

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume

Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange

Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

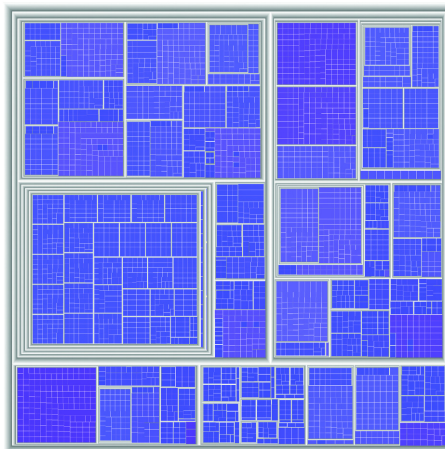
Containment:
Hierarchy Marks

Idiom	Treemaps
What: Data How: Encode Why: Tasks Scale	Tree. Area marks and containment, with rectilinear layout. Query attributes at leaf nodes. Leaf nodes: one million. Links: one million.



Idiom: Treemaps (cont.)

- Treemap layout showing hierarchical structure with containment rather than connection, in contrast to the node–link diagrams of the same 5161-node tree



Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

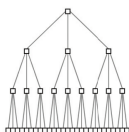
Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

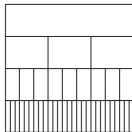
Visual encoding idioms for tree data



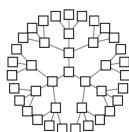
Seven visual encoding idioms showing the same tree dataset, using different combinations of visual channels. (a) Rectilinear vertical node-link, using connection to show link relationships, with vertical spatial position showing tree depth and horizontal spatial position showing sibling order. (b) Icicle, with vertical spatial position and size showing tree depth, and horizontal spatial position showing link relationships and sibling order. (c) Radial node-link, using connection to show link relationships, with radial depth spatial position showing tree depth and radial angular position showing sibling order. (d) Concentric circles, with radial depth spatial position and size showing tree depth and radial angular spatial position showing link relationships and sibling order. (e) Nested circles, using radial containment, with nesting level and size showing tree depth. (f) Treemap, using rectilinear containment, with nesting level and size showing tree depth. (g) Indented outline, with horizontal spatial position showing tree depth and link relationships and vertical spatial position showing sibling order



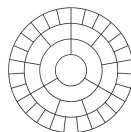
(a)



(b)



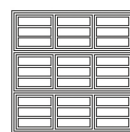
(c)



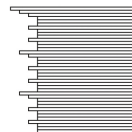
(d)



(e)



(f)



(g)



Example: GrouseFlocks

- **Compound network** is the combination of a network and tree on top of it, where the nodes in the network are the leaves of the tree.

Arrange Networks and Trees

Connection: Link
Marks

Matrix Views

Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

Idiom: GrouseFlocks



System	GrouseFlocks
What: Data	Network.
What: Derived	Cluster hierarchy atop original network.
What: Encode	Connection marks for original network, containment marks for cluster hierarchy.

Idiom: GrouseFlocks (cont.)

Geometry

Scalar Fields: One
Value

Isocontours

Direct Volume
Rendering

Vector Fields: Multiple
Values

Flow Glyphs

Geometric Flow

Texture Flow

Feature Flow

Tensor Fields: Many
Values

Arrange

Networks and
Trees

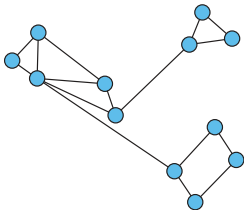
Connection: Link
Marks

Matrix Views

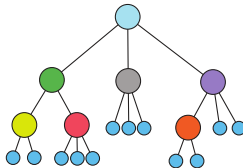
Costs and Benefits:
Connection versus
Matrix

Containment:
Hierarchy Marks

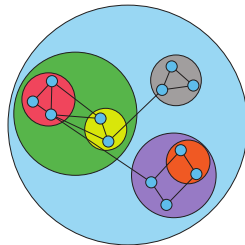
- GrouseFlocks uses containment to show graph hierarchy structure. (a) Original graph. (b) Cluster hierarchy built atop the graph, shown with a node– link layout. (c) Network encoded using connection, with hierarchy encoded using containment.



(a)



(b)



(c)

References



Goodfellow, I., Bengio, Y., and Courville, A. (2016).

Deep learning.

MIT press.



Munzner, T. (2014).

Visualization analysis and design.

CRC press.



Russell, S. and Norvig, P. (2016).

Artificial intelligence: a modern approach.

Pearson Education Limited.



Ward, M. O., Grinstein, G., and Keim, D. (2015).

Interactive data visualization: foundations, techniques, and applications.

CRC Press.